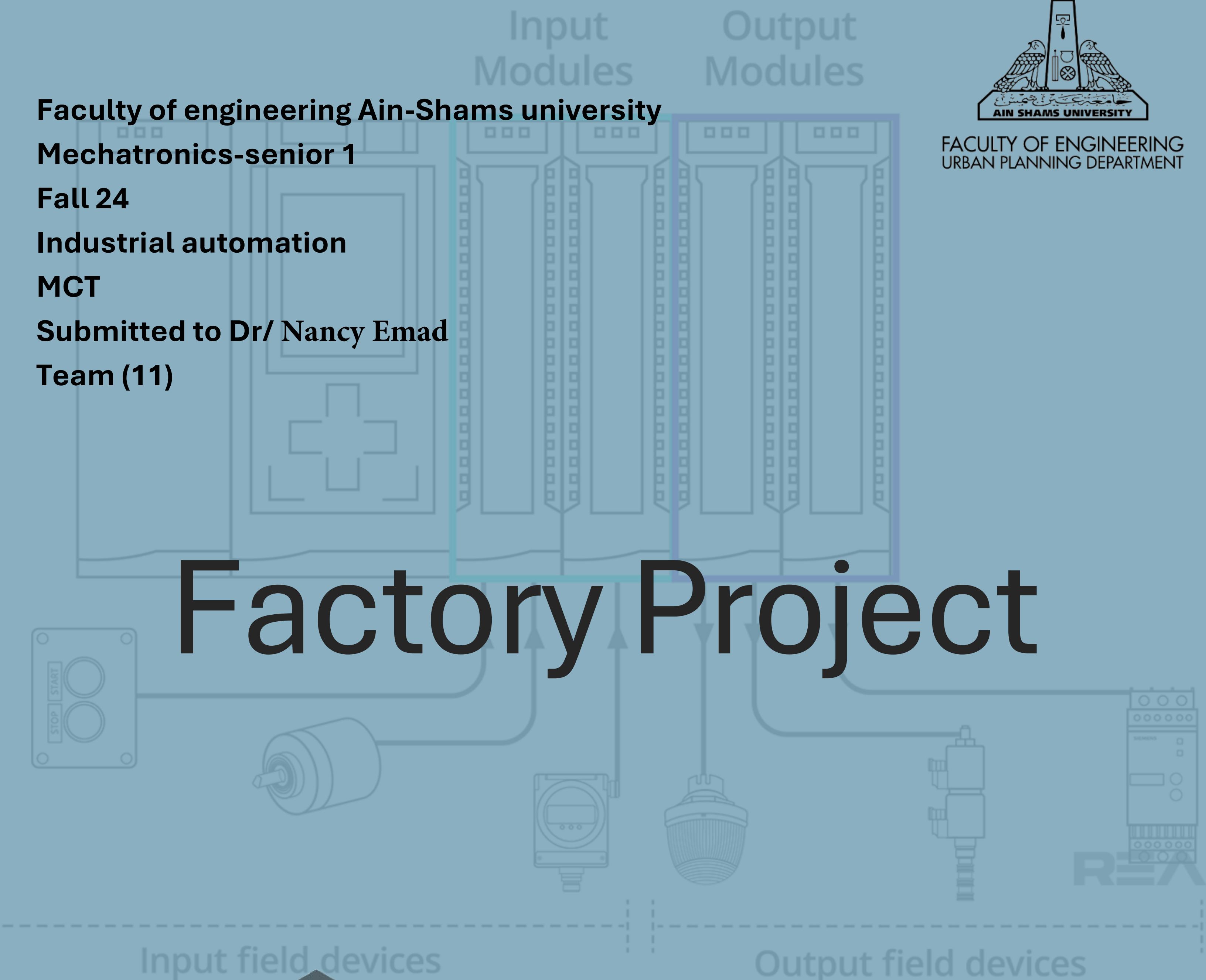
FACULTY OF ENGINEERING
URBAN PLANNING DEPARTMENT



Faculty of engineering Ain-Shams university
Mechatronics-senior 1
Fall 24
Industrial automation
MCT
Submitted to Dr/ Nancy Emad
Team (11)

Factory Project

Name	ID	Contribution
Adham Sabri	2100703	19%
Ahmed Ismail	2100722	15.5%
Hla Ehab	2100708	15.5%
Kholod Ahmed	2100712	16%
Suhaila Khaled	2001835	15.5%
Taha Gamal	2100566	18.5%

Table of Contents

0. List of figures and tables

NAME	TASKS
Adham Sabry	• Assembly station • HMI
Ahmed Ismail	• Sorting Blue or green
Hla Ehab	• Sorting Base or lid function
Kholoud Ahmed	• Sorting Base or lid function
Suhaila Khaled	• Technical report implementation
Taha Gamal	• CNC start and stop • stopping the station • factory io

Table 1 Task allocation

1. Abstract

Creating a production line containing stations: **Feeding**, through the conveyors, parts are fed to the pick and place machines. **sorting**, which sorts the raw material (blue or green). Through **the second sorting**, it separates one part for the lid manufacturing and one for the base manufacturing. **Machining**, getting 2 parts of each color to the CNC machines, total 4 machines to get out: green base, blue base, green lid and blue lid. And **assembly**, for assembling each lid to each base of each color. During that, the counter of the parts counts each assembled part, and when the desired number of parts are reached, the production line stops, however, the CNC will continue the part in it and after that it stops on the final conveyor without reaching the assembly station.

2. Introduction to TIA and Factory IO

a. TIA portal

The Totally Integrated Automation (TIA) Portal is designed to be an application that allows a programmer to design PLC programs, HMI displays, and motion/motor control systems, all from a single software.

Depending on the nature of the project and the software package purchased, not all of these systems will be visible in the immediate setup. For now, we'll focus only on the most basic target of the TIA portal which is the PLC and ladder diagram programming.

More flexible, faster, and more productive: Innovative simulation tools, seamlessly integrated engineering, and transparent plant operation work perfectly together in TIA Portal. The new options benefit system integrators and machine builders as well as plant operators, making TIA Portal your perfect gateway to automation in the Digital Enterprise.

I. Digital workflow

Increase your flexibility and work in an open, virtual, and networked manner

A digital workflow lets you use virtual representations of machines and systems. You can use these to simulate and test every aspect before actually building the real thing. This “digital twin” is also connected to operational IT and the Cloud for even more flexibility and ultimately resulting in higher quality products.

II. Integrated engineering

The heart of efficient automation

All important automation components are integrated into TIA Portal. This means that everything from controllers to energy management can be programmed. A consistent database and libraries with frequently used functions make engineering even faster and easier.

III. Transparent operation

Better overview of production processes, machine condition, and energy consumption

As global competition increases, it’s essential that you know exactly what’s happening in a machine. Transparent operation means that you can make the right decisions thanks to consistent data and in-depth analytics. This also helps you improve energy efficiency – an increasingly important factor in boosting your competitiveness.

b. Factory IO

Factory I/O is a 3D factory simulation for learning automation technologies. Designed to be easy to use, it allows to quickly build a virtual factory using a selection of common industrial parts. Factory I/O also includes many scenes inspired by typical industrial applications, ranging from beginner to advanced difficulty levels.

The most common scenario is to use Factory I/O as a PLC training platform since PLC are the most common controllers found in industrial applications. However, it can also be used with microcontrollers, SoftPLC, Modbus, among many other technologies.

I. I/O Drivers

An I/O Driver is a built-in feature of Factory I/O responsible for "talking" to an external controller. Factory I/O includes many I/O Drivers, each one intended to be used with a specific technology. You select a driver based on the controller you want to use.

II. Parts

Factory I/O provides a collection of parts based on the most common industrial equipment. These parts are organized into eight categories: items, heavy load parts, light load parts, sensors, operators, stations, warning devices and walkways.

III. Scenes

Factory I/O includes a list of ready-to-use scenes inspired by typical industrial applications. These scenes can be seen as ranging from beginner to advanced difficulty levels. However, a difficulty level is an abstract concept since it will depend on how far you want to take the design of a control algorithm. Note that these scenes are read-only and cannot be overwritten. In order to customize one of these scenes, save and open it from **My Scenes**.

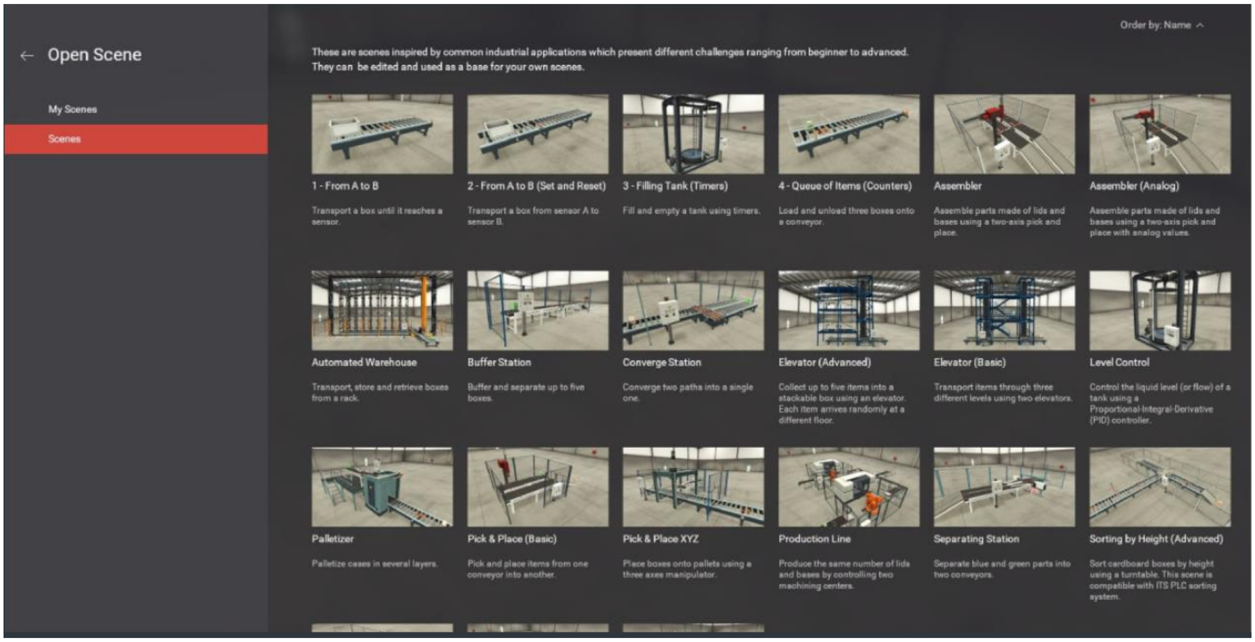


Figure 1 scenes provided by factory io

3. Whole Production line

Operator:

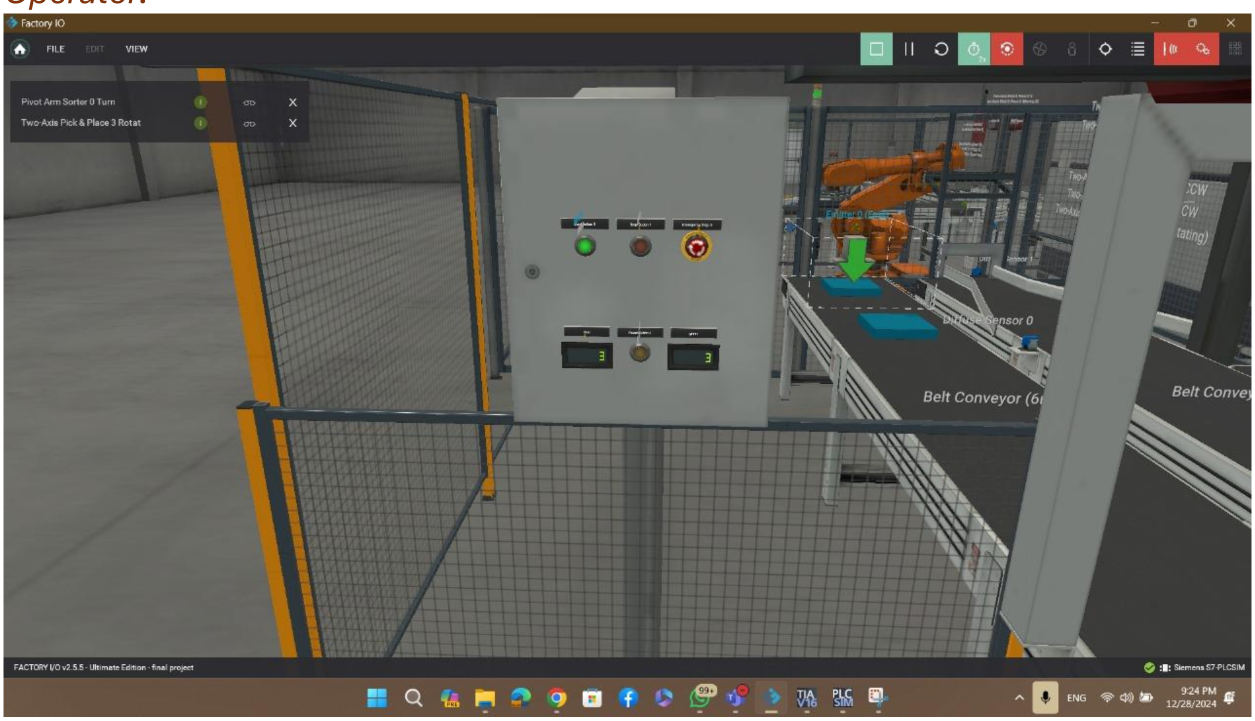


Figure 2 control panel in factory io

Figure 3 the whole production line using Factory I/O

Figure 4 production line

4. Feeding and sorting units

I. Parts

a. Emitter

Emits an Item to be used in a scene (e.g. cardboard box, pallet, etc.). While an item is still within the emitter volume, no more items are emitted. You can choose which part or base to emit, the time between emissions, the number of items to emit and whether random position and/or orientation should be taken into account. An emitter can be enabled or disabled by switching its tag On or Off.

Figure 5 emitter in factory io

6	0000 0000 0010 0000	Green raw material
7	0000 0000 0100 0000	Metal raw material

Figure 6 part to emit in factory io

The number of items to emit. If set to zero, it will emit up to the maximum number of items allowed in a scene (500).

Random Part Position/Orientation

Each emitted part will appear at a random position and/or orientation within the volume.

b. Remover

Removes one or more Items from the scene (e.g. cardboard box, pallet, product lid) when they intersect the remover's volume. You can enable or disable a remover by switching its tag on/off.

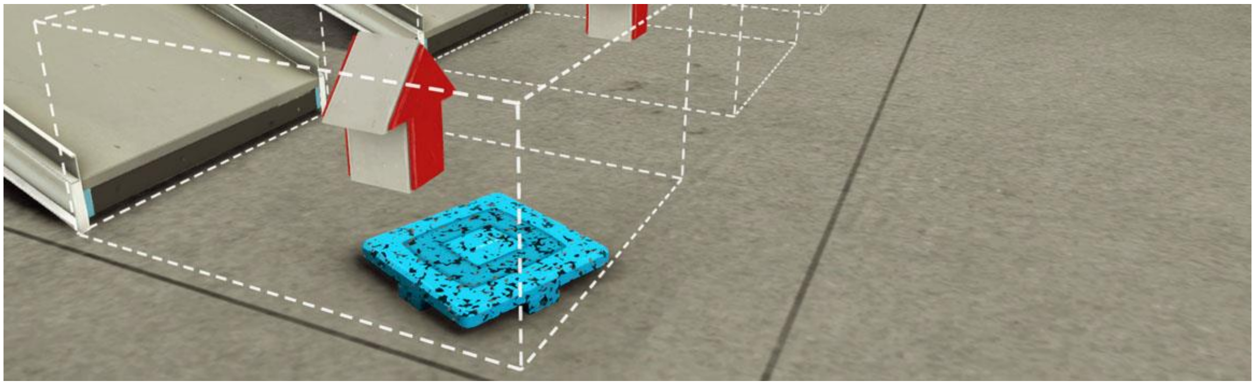


Figure 7 remover in factory io

c. Sensors

Vision sensor: The Vision Sensor recognizes [Raw Materials], [Product Lids] and [Product Bases], and their respective colors.

- LED: red (detecting)
- Detectable materials: Raw Materials, Product Bases, Product Lids
- Sensing range: 0.3 - 2 m

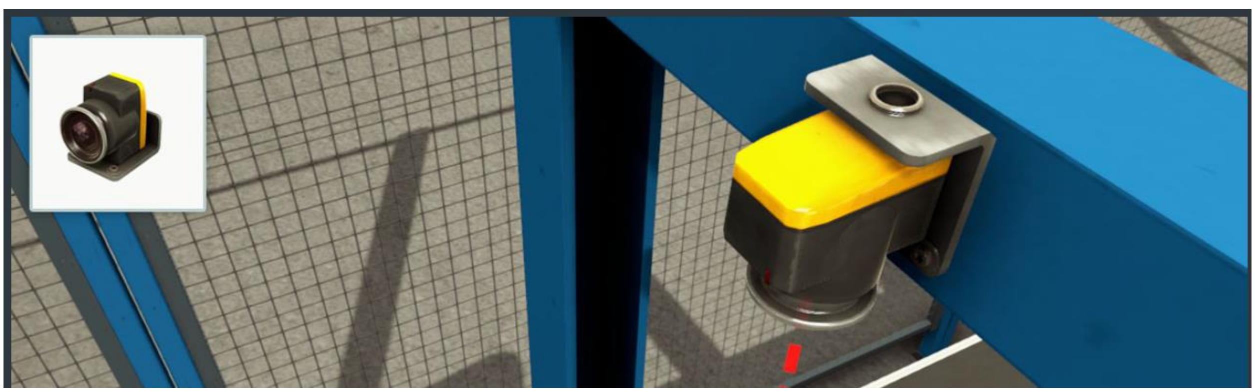


Figure 8 vision sensor in factory io

Item	All Digital	All Numerical	All ID
	Bit 0 1 2 3	*Value*	*Value*
None	0 0 0 0	0	0
Blue Raw Material	1 0 0 0	1	ID
Green Raw Material	0 0 1 0	4	ID

Figure 9 Items Encoding Per Configuration for vision sensor in factory io

d. Light load parts

1. **Belt conveyors**: Belt Conveyors are used to transport light load cargo. Can be controlled by digital or analog values.

    - Available lengths: 2, 4 and 6 m
    - Maximum conveying speed: 0.6 m/s (digital); 3 m/s (analog)

Figure 10 belt conveyor in factory io

2. **Curved Belt Conveyor**

Figure 11 Curved Belt Conveyor in factory io

3. **Straight Spur Conveyor**: Infeed/outfeed multi-belt conveyor, commonly used for accurate merging of loads into belt conveyors. Digital and analog values can control the conveyor.
    - Maximum conveying speed: 0.8 m/s (digital); 3 m/s (analog)

Figure 12 Straight Spur Conveyor in factory io

<u>Number of parts used</u>: 2

4. **Pivot Arm Sorter:** A 45º power face arm diverter, powered by a gearmotor. Equipped with a belt that helps to deviate the conveyed items onto the next part. The arm can rotate left or right according to the selected configuration.
    - Belt speed: 2 m/s

- Arm angular speed: 5 rad/s

Figure 13 Pivot Arm Sorter

5. **Aligners:** Thin metal structures, attachable to a conveyor to prevent parts from falling when transported at high speeds. There are four types of aligners available and in several colors**.**

Figure 14 Aligners

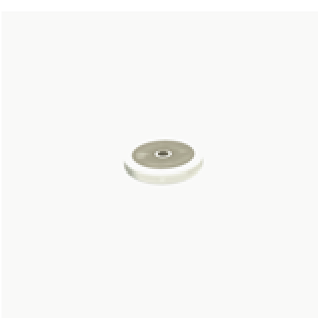


Figure 15 wheel aligner

6. **Chute Conveyor**: Straight chute conveyor, mostly used to dispatch items from belt conveyors.

Figure 16 Chute Conveyor in factory io

7. **Bracket and Metal Corner:** The bracket is a metal structure used as a height barrier and can be used to attach sensors. The metal corner can be used for several purposes including sensor placement.

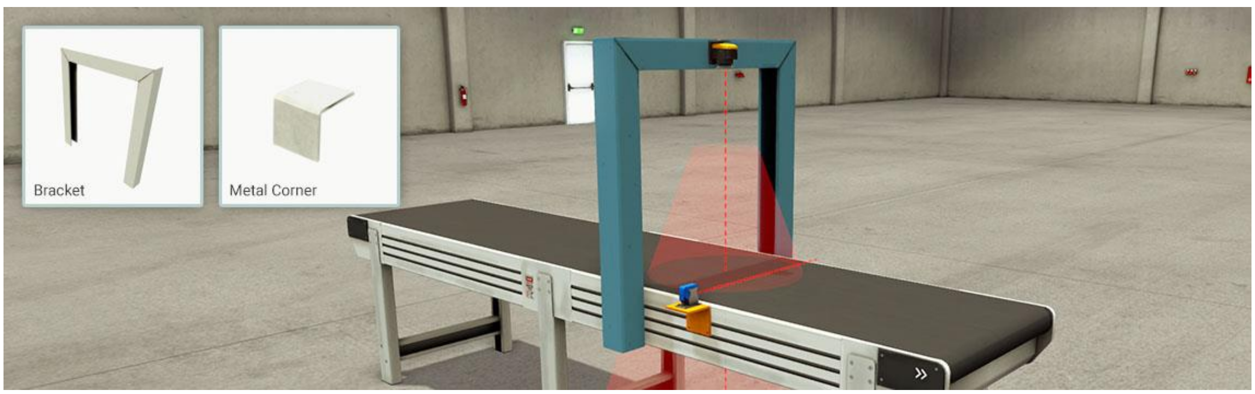


Figure 17 Bracket and Metal Corner factory io

e. Two-Axis Pick & Place

This part can be used to assemble Lids on Bases or pick and place items from one place to another. To guarantee a correct fit, bases and lids should be properly aligned by Positioning Bars.

- X-axis stroke: 1.125 m
- Z-axis stroke: 0.625 m
- Arm and picker speed: 2 m/s
- Z-axis rotation in 90° increments (arm and gripper)

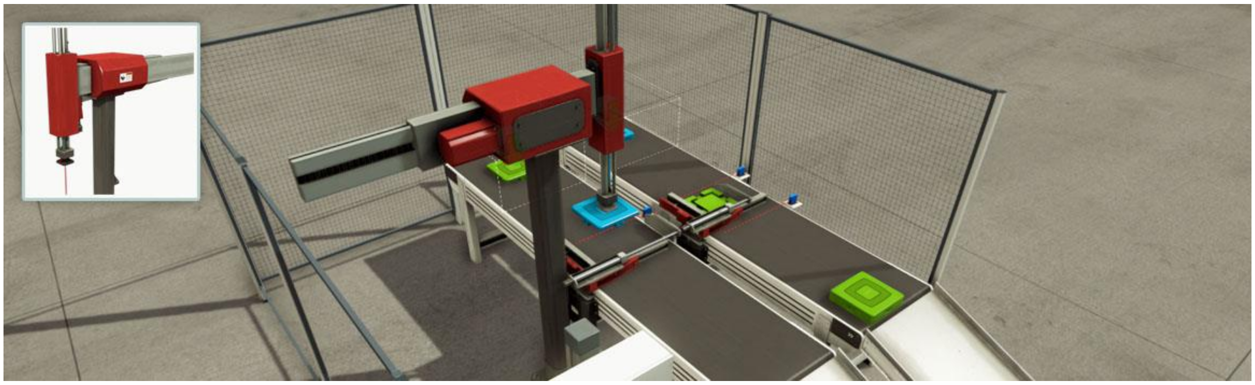


Figure 18 Two-Axis Pick & Place in factory io

*<u> Number of units used: </u>*3.

f. Items

Figure 20 green raw material

Figure 19 blue raw material

g. Safeguard

Figure 21 Safeguard

II. Operation

Feeding the raw material through the conveyors, getting sorted by the sensors, reaching the machining center.

III. Algorithm

When start is pressed:

- The entry conveyors start working, carrying the parts.
- When the part reaches the sensor, it detects the color.
- The blue parts would continue on the same way
- The blue part goes to the first CNC machine, manufacturing a blue base.
- The green parts would be picked up by the pick and place and placed on another conveyor.
- That second conveyor would take the part to the second CNC machine, manufacturing the green base

When a second green part is fed:

- The pick and place would place it in a third conveyor
- The third conveyor would take the part to the third CNC machine, manufacturing the green lid.

When a second blue part is fed:

- The pick and place would place it in a fourth conveyor
- The fourth conveyor would take the part to the fourth CNC machine, manufacturing the blue lid.

This operation continues until the desired number of parts is reached, after that it will use the pivot arm to push the extra parts to storage.

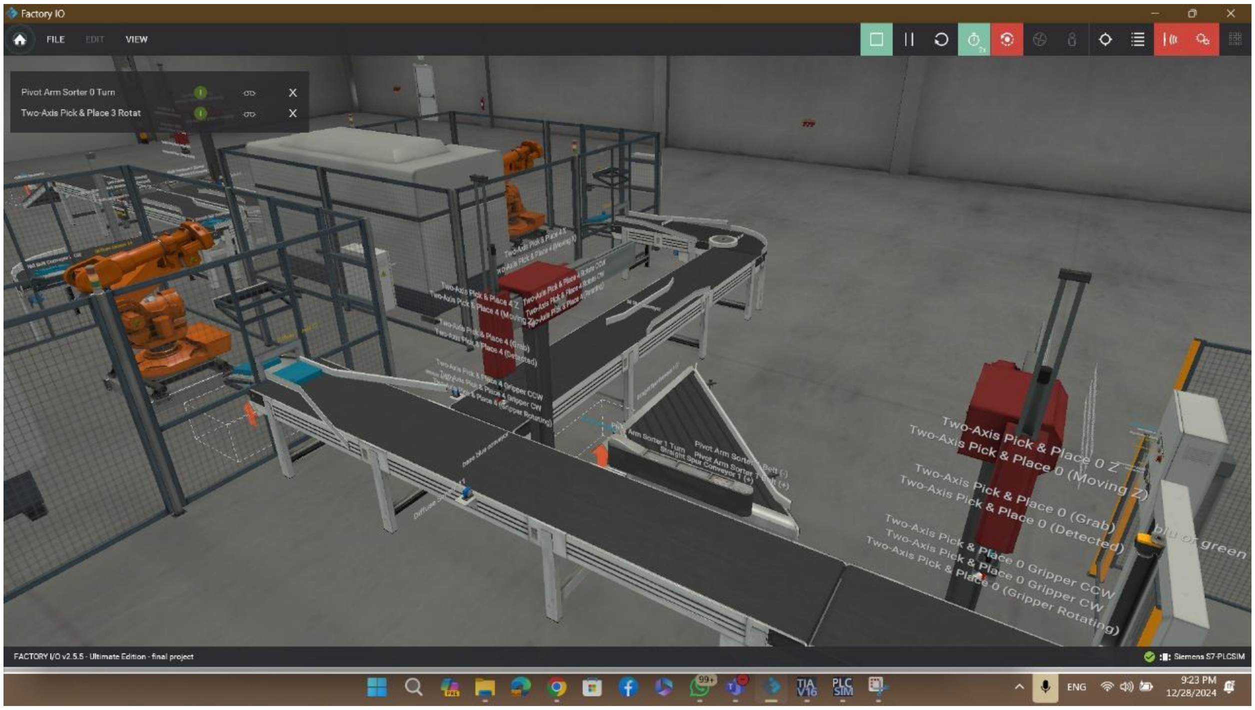


Figure 22 feeding conveyors

Figure 23 feeding conveyors II

5. Manufacturing

I. Parts

a. Sensors

Diffuse sensor: Diffuse photoelectric sensor which can detect any solid object.

- LED: red (detecting)
- Detectable materials: solids
- Sensing range: 0 - 1.6 m

Figure 24 diffuse sensor

b. Stations

Machining Center: The Machining Center is a station used to manufacture lids and bases from raw materials. First, the articulated robot waits for raw material to be placed at the entry bay. When new material is detected it is loaded into the CNC machine, which will start manufacturing an item. Each item type takes a different interval of time to be produced (lids: 6 seconds; bases: 3 seconds). Once the operation is complete, the robot places the item on the exit bay.

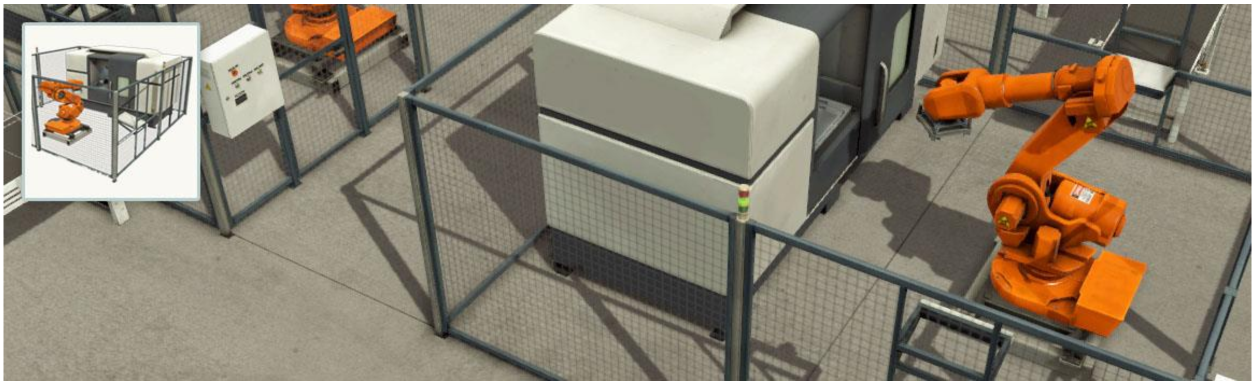


Figure 25 Machining Center

c. Light load parts

1. **Belt conveyor**
2. **Curved belt conveyors**
3. **Aligners**

II. Operation

Detecting the presence of a part, the arm takes the part to the CNC.

III. Algorithm

When the sensor detects the presence of a part:

- The arm moves and carries the part to the CNC
- The CNC starts working

When the CNC finishes:

- The arm takes the part to the next station (assembly)

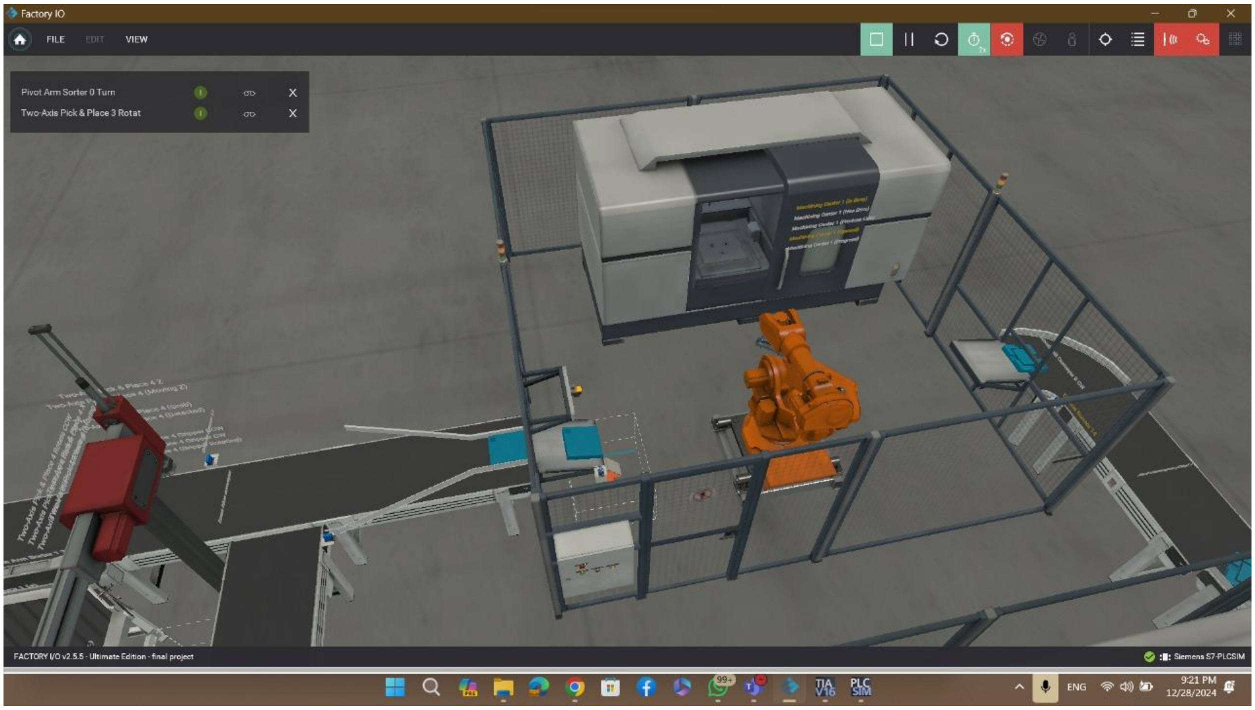


Figure 26 CNC machines

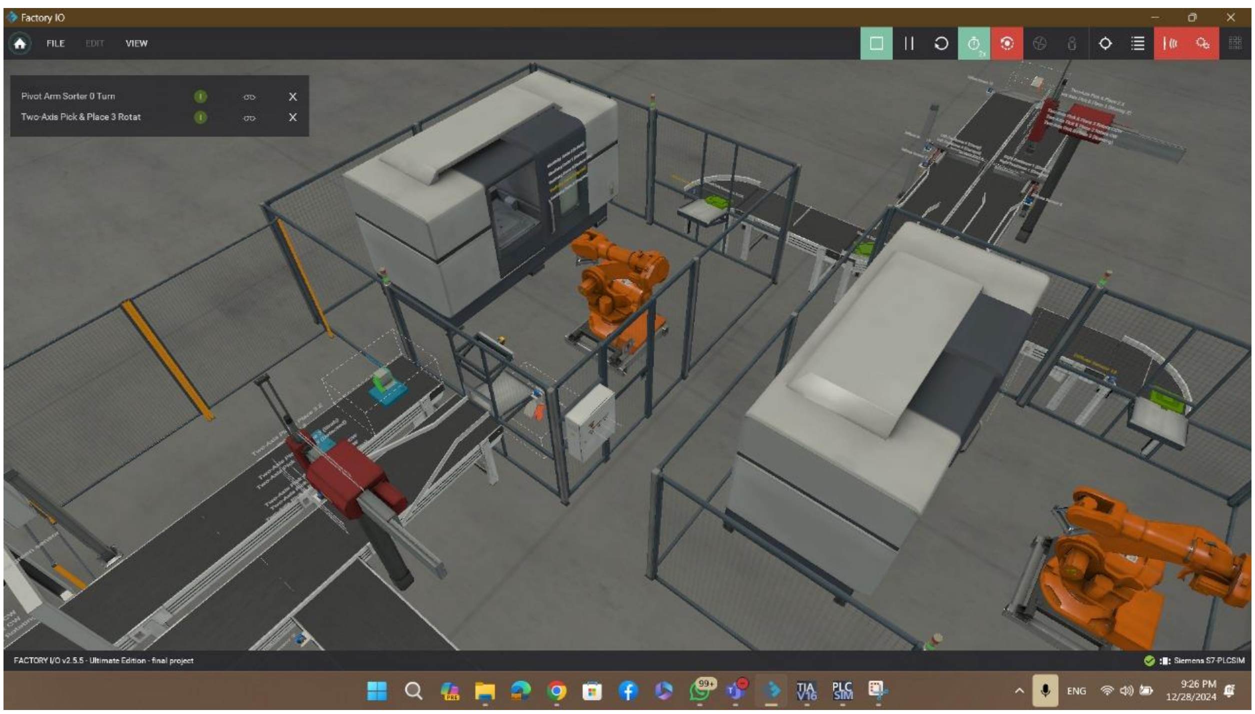


Figure 27 CNC machines II

6. Assembly unit

I. Parts

a. Sensors

Diffuse sensor

b. Light load parts

1. **Belt conveyors**
2. **Aligners**
3. **Positioning Bars :** A device used to consecutively position items at the same place by clamping them. Available in two different configurations, left and right.

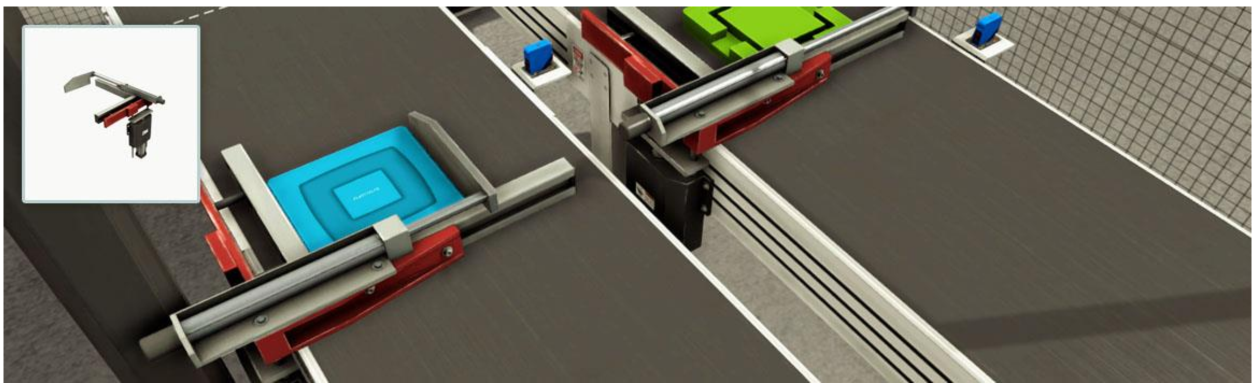


Figure 28 Positioning Bars in factory Io

Number of parts used: 2.

4. **Chute conveyors**

c. Stations

Two-Axis Pick & Place

II. Operation

Receiving the parts from the CNC machines. The first produced part will always be a base, the clamp would hold the base until the lid is ready, and then the pick and place would place the lid on top of the base.

III. Algorithm

When the base is finished:

- The exit conveyor would work, carrying the base.
- The sensor detects that the base has reached the desired position for the assembly.
- The clamp would clamp the part, waiting for the other part.

When the lid is finished:

- The conveyor would work, carrying the lid.
- The sensor detects that the lid has reached the desired position for assembly.
- The clamp would fix the position of the lid to be exactly like the base.
- The pick and place would pick the lid and then place it on top of the base.
- The conveyor must work immediately after the machine lefts the lid on top of the base
- However, there is a small delay, not to let the assembled part fall due to the moving conveyor and the rising of the clamp, then the exit conveyor would work.
- The counter counts the assembled parts.

Figure 29 assembly station

7. HMI

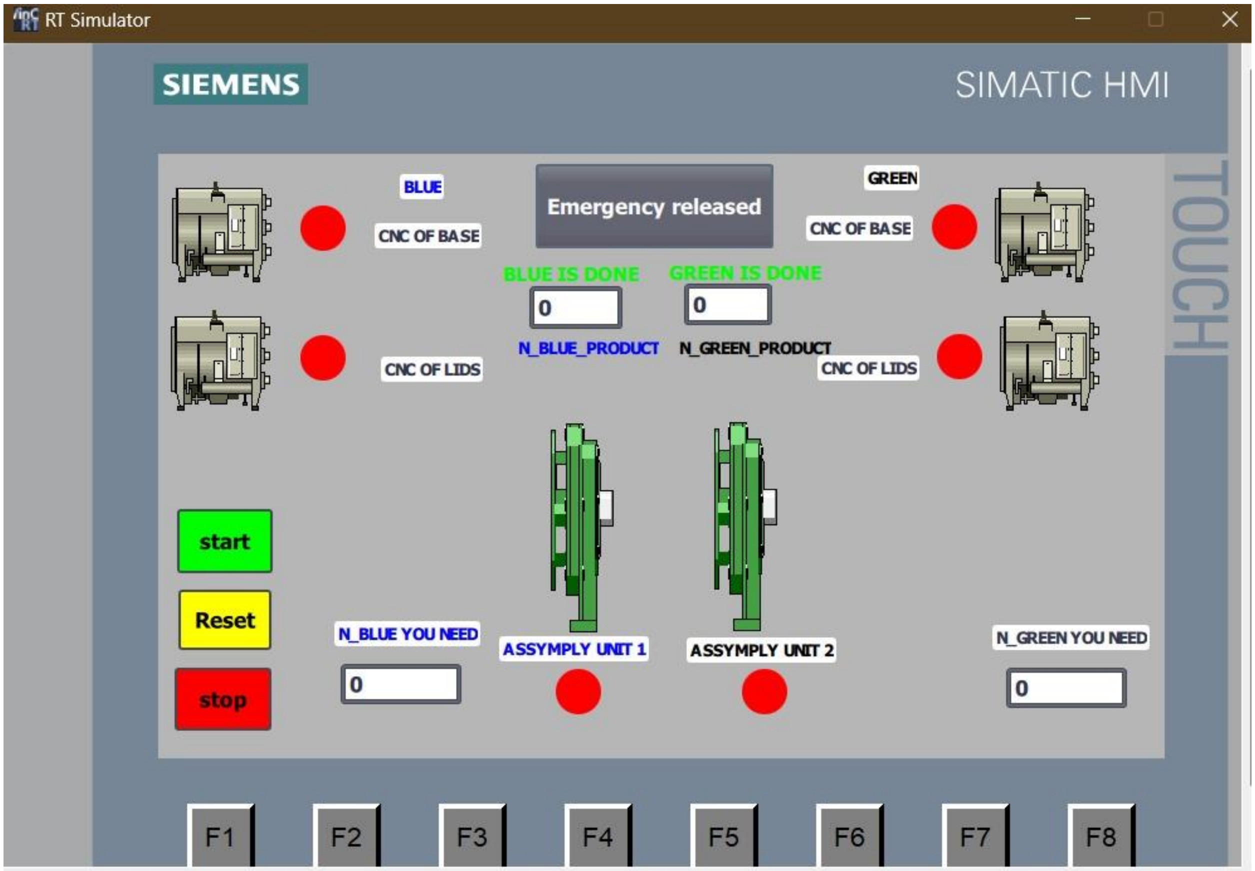


Figure 30 HMI

8. DRIVERS

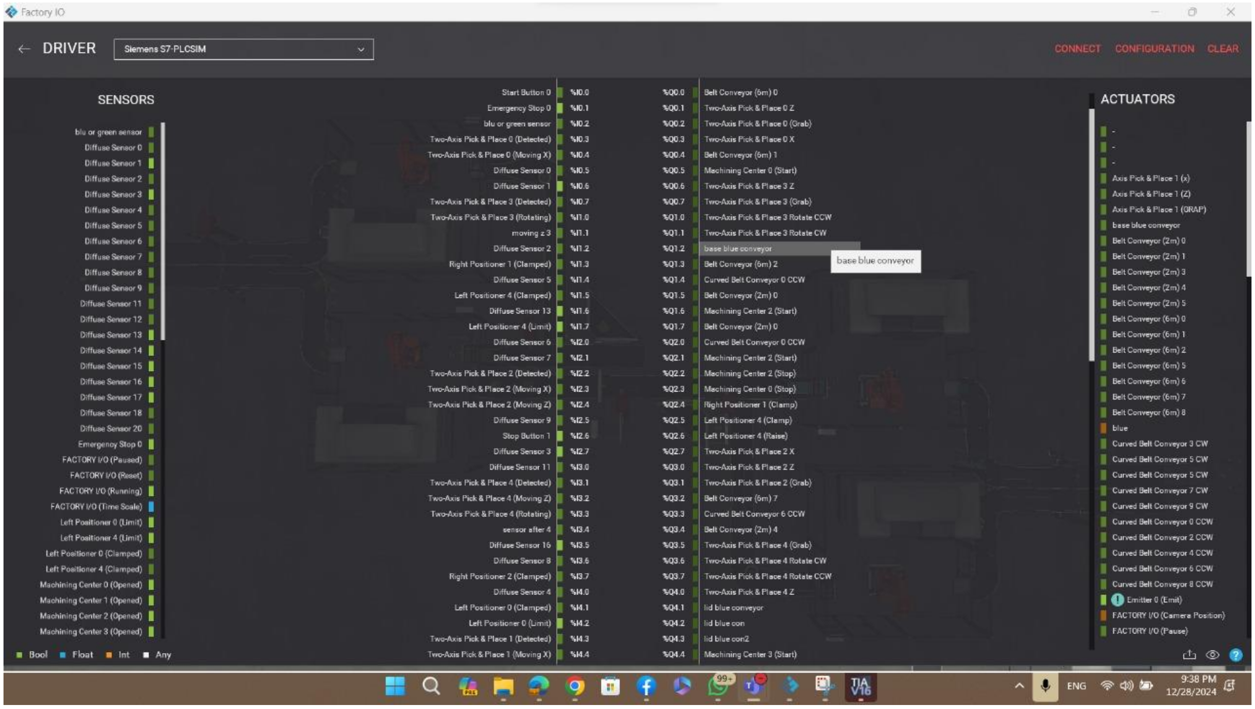


Figure 31 drivers communicating with TIA

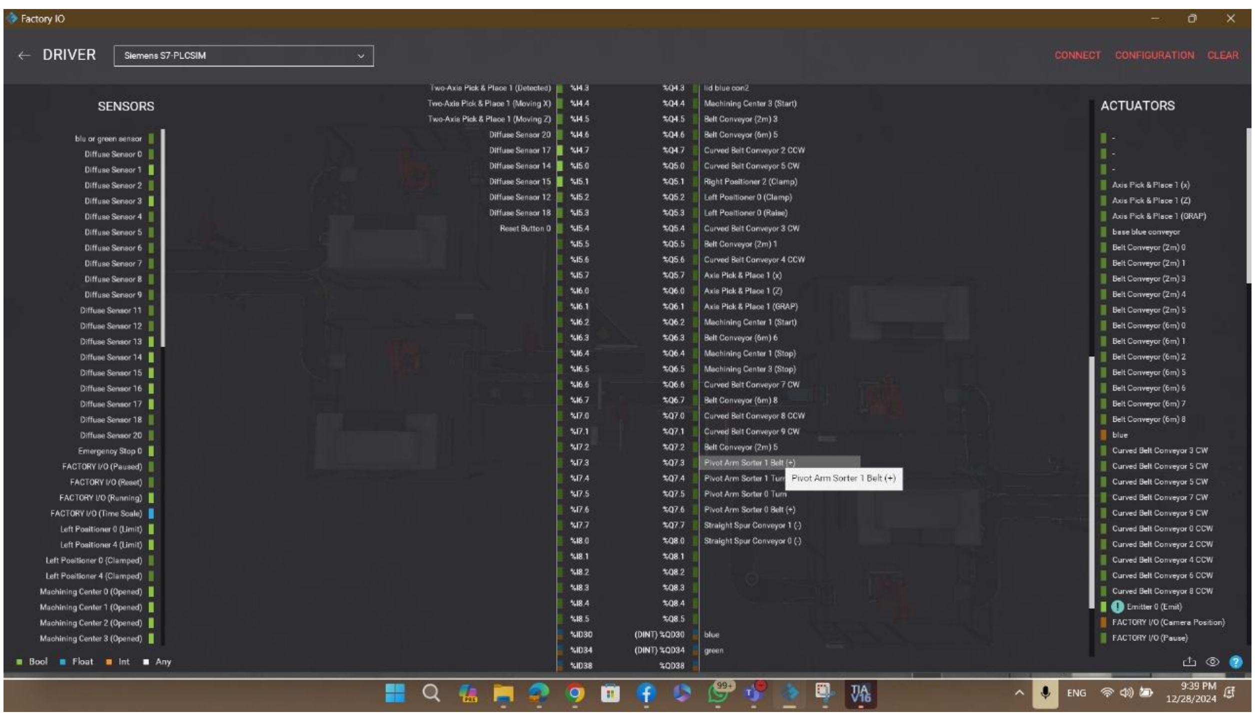


Figure 32 drivers communicating with TIA II

9. Drive link

https://drive.google.com/drive/folders/1jWPdvPN05TNjpQz9lUA599fIpiwGh2iC?usp=drive_link

10. Refs

- https://docs.factoryio.com/
- https://control.com/technical-articles/intro-to-siemens-s7-1200-plc-and-tia-portal-programming/#:~:text=The%20Totally%20Integrated%20Automation%20(TIA,all%20from%20a%20single%20software.
- https://www.siemens.com/global/en/products/automation/industry-software/automation-software/tia-portal.html
- https://docs.factoryio.com/manual/parts/sensors/#vision-sensor
- https://docs.factoryio.com/manual/parts/emitter/